CSE121:ORIENTATION TO COMPUTING-II

L:2 T:0 P:0 Credits:2

Course Outcomes: Through this course students should be able to

CO1 :: Understand the need for data science and big data along with its tools, job roles, and skill set

CO2 :: Discuss the use of AI and machine learning in different fields along with its tools, job roles, and skill set

CO3 :: Recognize the attacks, malware, tools, job roles, suspicious links, secure connection and skill set for cyber security

CO4 :: Recognize various DevOps, Software testing tools with their applications, along with their job roles and skill sets.

CO5 :: Discuss different types of cloud model implementation, services along with their tools, job roles, and skill set.

CO6 :: Understand the concepts of front-end, back-end, and AR/VR technologies along with their tools, job roles, and required skill sets.

Unit I

Data Science & Big Data : Data science and its need, Applications of data science/Big data, Data science Lifecycle with use case, Big data and its 3Vs, Challenges of Big data, Skill needed for Big data, Tools usage like Apache Hadoop, Tableau, R language, Excel, Big Data on the Cloud, Use of Big Data in different areas, Job roles and skillset for Data science and Big data

Unit II

Artificial Intelligence & Machine Learning : Introduction to AI, ML and Deep Learning, Expert systems, Fuzzy systems, Augmented Reality, Use of AI in different fields - NLP, Healthcare, Agriculture, Social media monitoring, Tools and techniques for implementing AI, Google Translator, Driverless Car, ALEXA, Siri, ChatGPT, Current trends and opportunities, Job roles and skillset for AI and ML

Unit III

Cybersecurity : Introduction to cybersecurity-definition, importance in digital era, CIA Triad (Confidentiality, Integrity, Availability), Cyber Threat Landscape: Insider vs External threats, Malwares, Common cyber-attacks- Phishing and Social Engineering attacks,Password and brute-force attacks Malware-based attacks, Denial of Service (DoS),Zero day attack, case study of recent cyber incidents, secure web browsing, social media and email security, personal data protection and digital footprints, Cybersecurity tools (Nmap, Wireshark, AI-based threat detection systems), cybersecurity compliance, Job roles and skill sets for cybersecurity

Unit IV

DevOps & Software Testing : Introduction to DevOps, DevOps Vs Traditional Software Development Models, DevOps Tools - Git, Docker, Selenium, Mavin, Puppet, Ansible, Kubernetes, Nagios, DevOps life cycle, role of automation in DevOps, CI/CD, fundamentals of testing, Objectives of Testing, Types of Testing, Levels of testing, Applications of software testing in IT companies, Manual Vs automation testing, Introduction to test case design (simple example), defect life cycle, Career opportunities in the field of DevOps and software testing with skillset

Unit V

Cloud Computing : Introduction to cloud computing, Uses of cloud computing in applications services, Platform deployments, Types of cloud model implementations, Types of cloud services, Data analytics, Virtualization, Tools and techniques for implementing cloud computing, Job roles and skillset for cloud computing

Unit VI

Full Stack Web Development & UI/UX : Introduction to Web Development, User Interface Design, frontend, backend, databases, CRUD applications, Responsive web design, mobile-first development, Single page applications (SPA), Languages such as HTML, CSS, PHP, Java Scripts, and frameworks by using VS code tool, Overview of AR/VR-significance, How does AR/VR work? Software/Hardware and devices, Applications, Job-roles and skillset for full stack and UI/UX

References:

1. BIG DATA by ANIL MAHESHWARI, MC GRAW HILL
2. PRINCIPLES OF SOFT COMPUTING by S.N. SIVANANDAM, S.N. DEEPA, WILEY

References:

3. HTML, CSS, AND JAVASCRIPT ALL IN ONE, SAMS TEACH YOURSELF by JULIE C. MELONI, JENNIFER KYRNIN, PEARSON

4. CLOUD COMPUTING: FUNDAMENTALS, INDUSTRY APPROACH AND TRENDS by RISHABH SHARMA, WILEY